

$$A_\mu \rightarrow A'_\mu = A_\mu - \partial_\mu \chi,$$

and the new mass term becomes

$$\frac{1}{2}m_\gamma^2 A_\mu A^\mu \rightarrow \frac{1}{2}m_\gamma^2 (A_\mu - \partial_\mu \chi)(A^\mu - \partial^\mu \chi) \neq \frac{1}{2}m_\gamma^2 A_\mu A^\mu,$$

from which it is clear that the photon mass term is not gauge invariant. Hence the required U(1) local gauge symmetry can only be satisfied if the gauge boson of an interaction is *massless*. This restriction is not limited to the U(1) local gauge symmetry of QED, it also applies to the SU(2)_L and SU(3) gauge symmetries of the weak interaction and QCD. Whilst the local gauge principle provides an elegant route to describing the nature of the observed interactions, it works only for massless gauge bosons. This is not a problem for QED and QCD where the gauge bosons are massless, but it is in apparent contradiction with the observation of the large masses of W and Z bosons.

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The problem with particle masses is not restricted to the gauge bosons. Writing the electron spinor field as $\psi = e$, the electron mass term in QED Lagrangian can be written in terms of the chiral particle states as

$$\begin{aligned} -m_e \bar{e} e &= -m_e \bar{e} \left[\frac{1}{2}(1 - \gamma^5) + \frac{1}{2}(1 + \gamma^5) \right] e \\ &= -m_e \bar{e} \left[\frac{1}{2}(1 - \gamma^5) e_L + \frac{1}{2}(1 + \gamma^5) e_R \right] \quad (17.16) \\ &= -m_e (\bar{e}_R e_L + \bar{e}_L e_R). \end{aligned}$$

In the SU(2)_L gauge transformation of the weak interaction, left-handed particles transform as weak isospin doublets and right-handed particles as singlets, and therefore the mass term of (17.16) breaks the required gauge invariance.

17.5 The Higgs mechanism

In the Standard Model, particles acquire masses through their interactions with the Higgs field. In this section, the Higgs mechanism is developed in three distinct stages. First it is shown how mass terms for a scalar field can arise from a

broken symmetry. This mechanism is then extended to show how the mass of a gauge boson can be generated from a broken $U(1)$ local gauge symmetry. Finally, the full Higgs mechanism is developed by breaking the $SU(2)_L \times U(1)_Y$ local gauge symmetry of the electroweak sector of the Standard Model.

17.5.1 Interacting scalar fields

A Lagrangian consists of two parts, a kinetic term involving the derivatives of the fields and a potential term expressed in terms of the fields themselves. For example, in the Lagrangian of QED (17.15), the kinetic terms for the electron and photon are

$$i\bar{\psi}\gamma^\mu\partial_\mu\psi \quad \text{and} \quad -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}.$$

The potential term, which represents the interactions between the electron and photon fields, is

$$e\bar{\psi}\gamma^\mu\psi A_\mu.$$

This can be associated with the normal three-point interaction vertex of QED, shown on the left of Figure 17.4. In general, the nature of the interactions between the fields and the strength of the coupling is determined by the terms in the Lagrangian involving the combinations of the fields, here $\bar{\psi}\psi A$.

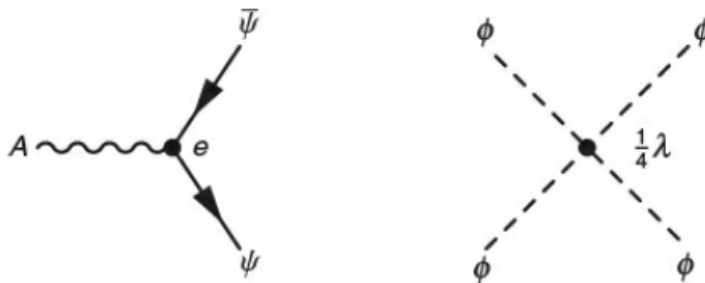


Fig. 17.4 The three-point interaction of QED and the four-point interaction for a scalar field with the potential $\lambda\phi^4/4$.

Now, consider a scalar field ϕ with the potential

$$V(\phi) = \frac{1}{2}\mu^2\phi^2 + \frac{1}{4}\lambda\phi^4. \quad (17.17)$$

The corresponding Lagrangian is given by

$$\begin{aligned} \mathcal{L} &= \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - V(\phi) \\ &= \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}\mu^2\phi^2 - \frac{1}{4}\lambda\phi^4. \end{aligned} \quad (17.18)$$

The term proportional to $(\partial_\mu\phi)(\partial^\mu\phi)$ can be associated with the kinetic energy of the scalar particle and $\frac{1}{2}\mu^2\phi^2$ can represent the mass of the particle. The ϕ^4 term can be identified as self-interactions of the scalar field, corresponding to the four-point interaction vertex shown in the right-hand plot of [Figure 17.4](#).

The vacuum state is the lowest energy state of the field ϕ and corresponds to the minimum of the potential of (17.17). For the potential to have a finite minimum, λ must be positive. If μ^2 is also chosen to be positive, the resulting potential, shown in [Figure 17.5a](#), has a minimum at $\phi = 0$. In this case, the vacuum state corresponds to the field ϕ being zero and the Lagrangian of (17.18) represents a scalar particle with mass μ and a four-point self-interaction term proportional to

ϕ^4 . However, whilst λ must be greater than zero for there to be a finite minimum, there is no such restriction for μ^2 . If $\mu^2 < 0$, the associated term in the Lagrangian can no longer be interpreted as a mass and the potential of (17.17) has minima at

$$\phi = \pm v = \pm \sqrt{\frac{-\mu^2}{\lambda}},$$

as shown in [Figure 17.5b](#). For $\mu^2 < 0$, the lowest energy state does not occur at $\phi = 0$ and the field is said to have a non-zero vacuum expectation value v . Since the potential is symmetric, there are two degenerate possible vacuum states. The actual vacuum state of the field either will be $\phi = +v$ or $\phi = -v$. The choice of the vacuum state breaks the symmetry of the Lagrangian, a process known as *spontaneous symmetry breaking*. A familiar example of spontaneous symmetry breaking is a ferromagnet with magnetisation $\mathbf{M}$. The Lagrangian (or Hamiltonian) depends on $\mathbf{M}^2$ and has no preferred direction. However, below the Curie temperature, the spins will be aligned in a particular direction, spontaneously breaking the underlying rotational symmetry of the Lagrangian.

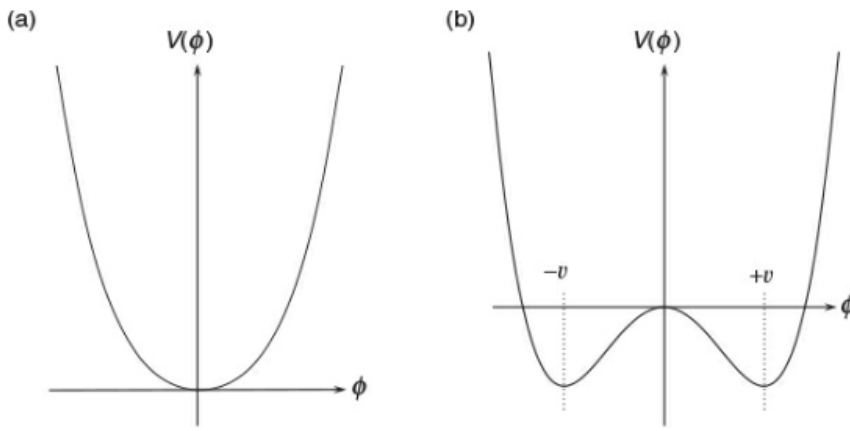


Fig. 17.5 The one-dimensional potential $V(\phi) = \mu^2\phi^2/2 + \lambda\phi^4/4$ for $\lambda > 0$ and the cases where (a) $\mu^2 > 0$ and (b) $\mu^2 < 0$.

If the vacuum state of the scalar field is chosen to be at $\phi = +v$, the excitations of the field, which describe the particle states, can be obtained by considering perturbations of the field ϕ around the vacuum state by writing $\phi(x) = v + \eta(x)$.

Since the vacuum expectation value v is a constant, $\partial_\mu \phi = \partial_\mu \eta$ and the Lagrangian of (17.18), expressed in terms of the field η , is

$$\begin{aligned}\mathcal{L}(\eta) &= \frac{1}{2}(\partial_\mu \eta)(\partial^\mu \eta) - V(\eta) \\ &= \frac{1}{2}(\partial_\mu \eta)(\partial^\mu \eta) - \frac{1}{2}\mu^2(v + \eta)^2 - \frac{1}{4}\lambda(v + \eta)^4.\end{aligned}$$

Since the minimum of the potential is given by $\mu^2 = -\lambda v^2$, this expression can be written as

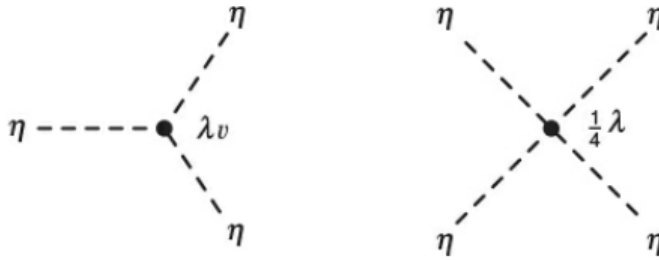
$$\mathcal{L}(\eta) = \frac{1}{2}(\partial_\mu \eta)(\partial^\mu \eta) - \lambda v^2 \eta^2 - \lambda v \eta^3 - \frac{1}{4}\lambda \eta^4 + \frac{1}{4}\lambda v^4. \quad (17.19)$$

From the comparison with the Lagrangian for a free scalar field of (17.5), it can be seen that the term proportional to η^2 can be interpreted as a mass

$$m_\eta = \sqrt{2\lambda v^2} = \sqrt{-2\mu^2},$$

and therefore the Lagrangian of (17.19) describes a massive scalar field. The terms proportional to η^3 and η^4 can be identified as triple and quartic interaction terms, as indicated in Figure 17.6. Finally, the term $\lambda v^4/4$ is just a constant, and has no physical consequences. Hence after spontaneous symmetry breaking, and having expanded the field about the vacuum state, the Lagrangian can be written as

$$\mathcal{L}(\eta) = \frac{1}{2}(\partial_\mu \eta)(\partial^\mu \eta) - \frac{1}{2}m_\eta^2 \eta^2 - V(\eta), \quad \text{with} \quad V(\eta) = \lambda v \eta^3 + \frac{1}{4}\lambda \eta^4. \quad (17.20)$$



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Fig. 17.6 The self-interactions of the field η that lead to Feynman diagrams for the processes $\eta \rightarrow \eta\eta$ and $\eta\eta \rightarrow \eta\eta$.

It is important to realise that the Lagrangian of (17.20) is the same as the original Lagrangian of (17.18), but is now expressed as excitations about the minimum at $\square = +v$. In principle, the same physical predictions can be obtained by using either form. However, in order to use perturbation theory, it is necessary to express the fields as small perturbations about the vacuum state.

17.5.2 Symmetry breaking for a complex scalar field

The idea of spontaneous symmetry breaking, introduced above in the context of a real scalar field, can be applied to the *complex* scalar field,

$$\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2),$$

for which the corresponding Lagrangian is

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$$\mathcal{L} = (\partial_\mu \phi)^* (\partial^\mu \phi) - V(\phi) \quad \text{with} \quad V(\phi) = \mu^2 (\phi^* \phi) + \lambda (\phi^* \phi)^2. \quad (17.21)$$

When expressed in terms of the two (real) scalar fields ϕ_1 and ϕ_2 this is just

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \phi_1) (\partial^\mu \phi_1) + \frac{1}{2} (\partial_\mu \phi_2) (\partial^\mu \phi_2) - \frac{1}{2} \mu^2 (\phi_1^2 + \phi_2^2) - \frac{1}{4} \lambda (\phi_1^2 + \phi_2^2)^2. \quad (17.22)$$

As before, for the potential to have a finite minimum, $\lambda > 0$. The Lagrangian of (17.21) is invariant under the transformation $\phi \rightarrow \phi' = e^{i\alpha} \phi$, because $\phi'^* \phi' = \phi^* \phi$, and therefore possesses a *global* U(1) symmetry. The shape of the potential depends on the sign of μ^2 , as shown in Figure 17.7. When $\mu^2 > 0$, the minimum of the potential occurs when both fields are zero. If $\mu^2 < 0$, the potential has an infinite set of minima defined by

$$\phi_1^2 + \phi_2^2 = \frac{-\mu^2}{\lambda} = v^2,$$

as indicated by the dashed circle in Figure 17.7. The physical vacuum state will correspond to a particular point on this circle, breaking the *global* U(1) symmetry

of the Lagrangian. Without loss of generality, the vacuum state can be chosen to be in the real direction, $(\phi_1, \phi_2) = (v, 0)$, and the complex scalar field ϕ can be expanded about the vacuum state by writing $\phi_1(x) = \eta(x) + v$ and $\phi_2(x) = \xi(x)$,

$$\phi = \frac{1}{\sqrt{2}} (\eta + v + i\xi).$$

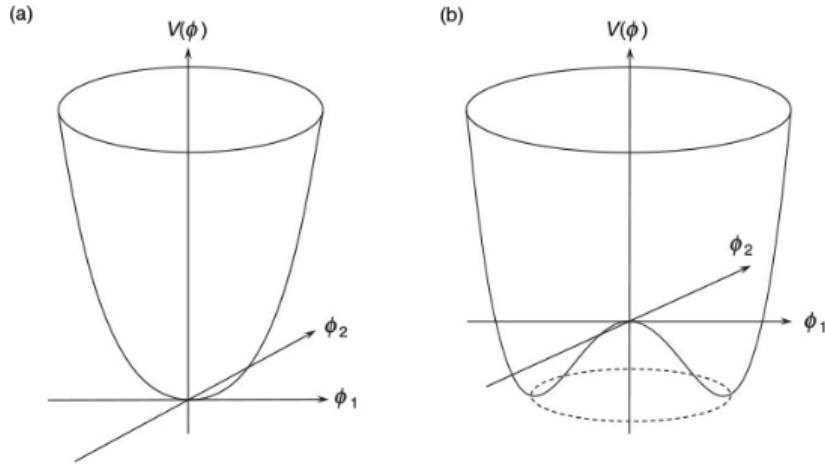


Fig. 17.7 The $V(\phi) = \mu^2(\phi^*\phi) + \lambda(\phi^*\phi)^2$ potential for a complex scalar field for (a) $\mu^2 > 0$ and (b) $\mu^2 < 0$.

The Lagrangian of (17.22), written in terms of the fields η and ξ , is

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \eta)(\partial^\mu \eta) + \frac{1}{2}(\partial_\mu \xi)(\partial^\mu \xi) - V(\eta, \xi),$$

where the potential $V(\eta, \xi)$ is given by

$$V(\eta, \xi) = \mu^2 \phi^2 + \lambda \phi^4 \quad \text{with} \quad \phi^2 = \phi \phi^* = \frac{1}{2} \left[(v + \eta)^2 + \xi^2 \right].$$

The potential can be written in terms of the fields η and ξ using $\mu^2 = -\lambda v^2$,

$$\begin{aligned} V(\eta, \xi) &= \mu^2 \phi^2 + \lambda \phi^4 \\ &= -\frac{1}{2} \lambda v^2 \left\{ (v + \eta)^2 + \xi^2 \right\} + \frac{1}{4} \lambda \left\{ (v + \eta)^2 + \xi^2 \right\}^2 \\ &= -\frac{1}{4} \lambda v^4 + \lambda v^2 \eta^2 + \lambda v \eta^3 + \frac{1}{4} \lambda \eta^4 + \frac{1}{4} \lambda \xi^4 + \lambda v \eta \xi^2 + \frac{1}{2} \lambda \eta^2 \xi^2. \end{aligned}$$

The term which is quadratic in the field η can be identified as a mass, and the terms with either three or four powers of the fields can be identified as interaction terms. Thus the Lagrangian can be written as

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \eta)(\partial^\mu \eta) - \frac{1}{2}m_\eta^2 \eta^2 + \frac{1}{2}(\partial_\mu \xi)(\partial^\mu \xi) - V_{int}(\eta, \xi), \quad (17.23)$$

with $m_\eta = \sqrt{2\lambda v^2}$ and interactions given by

$$V_{int}(\eta, \xi) = \lambda v \eta^3 + \frac{1}{4}\lambda \eta^4 + \frac{1}{4}\lambda \xi^4 + \lambda v \eta \xi^2 + \frac{1}{2}\lambda \eta^2 \xi^2. \quad (17.24)$$

These interaction terms correspond to triple and quartic couplings of the fields η and ξ , as shown in [Figure 17.8](#).



Fig. 17.8 The scalar interactions obtained by breaking the symmetry for a complex scalar field.

The Lagrangian of [\(17.23\)](#) represents a scalar field η with mass $m_\eta = \sqrt{2\lambda v^2}$ and a *massless* scalar field ξ . The excitations of the massive field η are in the direction where the potential is (to first order) quadratic. In contrast, the particles described by the massless scalar field ξ correspond to excitations in the direction where the potential does not change, as indicated in [Figure 17.9](#). This massless scalar particle is known as a Goldstone boson.

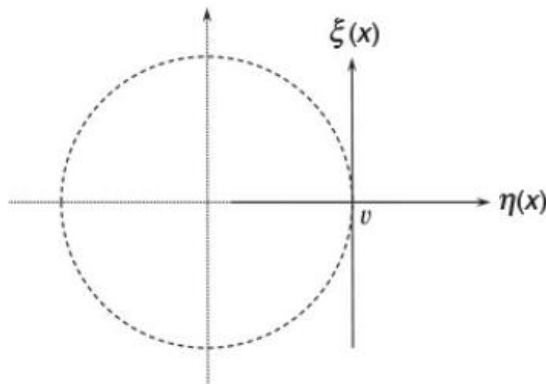


Fig. 17.9 The fields $\eta(x)$ and $\xi(x)$ in terms of the vacuum expectation value at $\square = (v, 0)$.

17.5.3 The Higgs mechanism

In the Higgs mechanism, the spontaneous symmetry breaking of a complex scalar field with a potential,

$$V(\phi) = \mu^2 \phi^2 + \lambda \phi^4, \quad (17.25)$$

is embedded in a theory with a *local* gauge symmetry. In this section, the example of a $U(1)$ local gauge symmetry is used to introduce the main ideas.

Because of the presence of the derivatives in (17.21), the Lagrangian for a complex scalar field ϕ is *not* invariant under the $U(1)$ *local* gauge transformation

$$\phi(x) \rightarrow \phi'(x) = e^{ig\chi(x)} \phi(x). \quad (17.26)$$

The required $U(1)$ local gauge invariance can be achieved by replacing the derivatives in the Lagrangian with the corresponding covariant derivatives

$$\partial_\mu \rightarrow D_\mu = \partial_\mu + igB_\mu.$$

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The resulting Lagrangian,

$$\mathcal{L} = (D_\mu \phi)^* (D^\mu \phi) - V(\phi^2),$$

is gauge invariant (see [Problem 17.7](#)) provided the new gauge field B_μ , which appears in the covariant derivative, transforms as

$$B_\mu \rightarrow B'_\mu = B_\mu - \partial_\mu \chi(x). \quad (17.27)$$

Just as was the case for Dirac Lagrangian (see [Section 17.3](#)), the required local gauge invariance implies the existence a new gauge field with well-defined gauge transformation properties. The combined Lagrangian for the complex scalar field ϕ and the gauge field B is

$$\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + (D_\mu \phi)^* (D^\mu \phi) - \mu^2 \phi^2 - \lambda \phi^4, \quad (17.28)$$

where $F^{\mu\nu} F_{\mu\nu}$ is the kinetic term for the new field with

$$F^{\mu\nu} = \partial^\mu B^\nu - \partial^\nu B^\mu.$$

The gauge field B is required to be massless, since the mass term $\frac{1}{2}m_B B_\mu B^\mu$ would break the gauge invariance. The term involving the covariant derivatives, when written out in full, is

$$\begin{aligned} (D_\mu \phi)^* (D^\mu \phi) &= (\partial_\mu - igB_\mu) \phi^* (\partial^\mu + igB^\mu) \phi \\ &= (\partial_\mu \phi)^* (\partial^\mu \phi) - igB_\mu \phi^* (\partial^\mu \phi) + ig(\partial_\mu \phi^*) B^\mu \phi + g^2 B_\mu B^\mu \phi^* \phi \end{aligned}$$

and the full expression for the Lagrangian is

$$\begin{aligned} \mathcal{L} = & -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + (\partial_\mu \phi)^* (\partial^\mu \phi) - \mu^2 \phi^2 - \lambda \phi^4 \\ & - igB_\mu \phi^* (\partial^\mu \phi) + ig(\partial_\mu \phi^*) B^\mu \phi + g^2 B_\mu B^\mu \phi^* \phi. \end{aligned} \quad (17.29)$$

For the case where the potential for the scalar field of (17.25) has $\mu^2 < 0$, the vacuum state is degenerate and the choice of the physical vacuum state spontaneously breaks the symmetry of the Lagrangian of (17.29). As before, the physical

vacuum state is chosen to be $\phi_1 + i\phi_2 = v$, and the complex scalar field ϕ is expanded about the vacuum state by writing

$$\phi(x) = \frac{1}{\sqrt{2}}(v + \eta(x) + i\xi(x)). \quad (17.30)$$

Substituting (17.30) into (17.29) leads to (see Problem 17.6)

$$\mathcal{L} = \underbrace{\frac{1}{2}(\partial_\mu \eta)(\partial^\mu \eta) - \lambda v^2 \eta^2}_{\text{massive } \eta} + \underbrace{\frac{1}{2}(\partial_\mu \xi)(\partial^\mu \xi)}_{\text{massless } \xi} - \underbrace{\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}g^2 v^2 B_\mu B^\mu}_{\text{massive gauge field}} - V_{\text{int}} + gvB_\mu(\partial^\mu \xi), \quad (17.31)$$

where $V_{\text{int}}(\eta, \xi, B)$ contains the three- and four-point interaction terms of the fields η , ξ and B . As before, the breaking of the symmetry of the Lagrangian produces a massive scalar field η and a massless Goldstone boson ξ . In addition, the previously massless gauge field B has acquired a mass term $\frac{1}{2}g^2 v^2 B_\mu B^\mu$, achieving the aim of giving a mass to the gauge boson of the local gauge symmetry. Again it should be emphasised that this is exactly the same Lagrangian as (17.28), but with the complex scalar field expanded about the vacuum state at $\phi_1 + i\phi_2 = v$; by expanding the scalar fields about the vacuum where the fields have a non-zero vacuum expectation value, the underlying gauge symmetry of the Lagrangian has been hidden, but has not been removed.

However, there appear to be two problems with (17.31). The original Lagrangian contained four degrees of freedom, one for each of the scalar fields η_1 and η_2 , and the two transverse polarisation states for the massless gauge field B . In the Lagrangian of (17.31), the gauge boson has become massive and therefore has the additional longitudinal polarisation state; somehow in the process of spontaneous symmetry breaking an additional degree of freedom appears to have been introduced. Furthermore, the $gvB_\mu(\partial^\mu\xi)$ term appears to represent a direct coupling between the Goldstone field ξ and the gauge field B . It would appear that the spin-1 gauge field can transform into a spin-0 scalar field, as indicated in Figure 17.10. This term is somewhat reminiscent of the off-diagonal mass term encountered in the discussion of the neutral kaon system, which coupled the K^0 and $\bar{K}^0$ flavour states, suggesting that the fields appearing in (17.31) are not the physical fields. This coupling to the Goldstone field ultimately will be associated with the longitudinal polarisation state of the massive gauge boson.

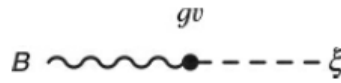


Fig. 17.10 The coupling between the gauge field B and the Goldstone field ξ .

The Goldstone field ξ in (17.31) can be eliminated from the Lagrangian by making the appropriate gauge transformation. By writing

$$\frac{1}{2}(\partial_\mu\xi)(\partial^\mu\xi) + gvB_\mu(\partial^\mu\xi) + \frac{1}{2}g^2v^2B_\mu B^\mu = \frac{1}{2}g^2v^2\left[B_\mu + \frac{1}{gv}(\partial_\mu\xi)\right]^2,$$

and making the gauge transformation,

$$B_\mu(x) \rightarrow B'_\mu(x) = B_\mu(x) + \frac{1}{gv}\partial_\mu\xi(x), \quad (17.32)$$

the Lagrangian of (17.31) becomes

$$\mathcal{L} = \underbrace{\frac{1}{2}(\partial^\mu\eta)(\partial_\mu\eta) - \lambda v^2\eta^2}_{\text{massive } \eta} + \underbrace{-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}g^2v^2B'^{\mu'}B'_{\mu'}}_{\text{massive gauge field}} - V_{int}.$$

Since the original Lagrangian was constructed to be invariant under local $U(1)$ gauge transformations, the physical predictions of the theory are unchanged by the gauge transformation of (17.32).

Thus, with the appropriate choice of gauge, the Goldstone field ξ no longer appears in the Lagrangian. This choice of gauge corresponds to taking $\chi(x) = -\xi(x)/gv$ in (17.27). The corresponding gauge transformation of the original complex scalar field $\phi(x)$ is therefore

$$\phi(x) \rightarrow \phi'(x) = e^{-ig\frac{\xi(x)}{gv}} \phi(x) = e^{-i\xi(x)/v} \phi(x). \quad (17.33)$$

After symmetry breaking, the complex scalar field was expanded about the physical vacuum by writing $\phi(x) = \frac{1}{\sqrt{2}}(v + \eta(x) + i\xi(x))$, which to first order in the fields can be expressed as

$$\phi(x) \approx \frac{1}{\sqrt{2}} [v + \eta(x)] e^{i\xi(x)/v}.$$

The effect of the gauge transformation of (17.33) on the original complex scalar field is therefore

$$\phi(x) \rightarrow \phi'(x) = \frac{1}{\sqrt{2}} e^{-i\xi(x)/v} [v + \eta(x)] e^{i\xi(x)/v} = \frac{1}{\sqrt{2}} (v + \eta(x)).$$

Hence, the gauge in which the Goldstone field $\xi(x)$ is eliminated from the Lagrangian, which is known as the *Unitary gauge*, corresponds to choosing the complex scalar field $\phi(x)$ to be entirely real,

$$\phi(x) = \frac{1}{\sqrt{2}} (v + \eta(x)) \equiv \frac{1}{\sqrt{2}} (v + h(x)).$$

Here the field $\eta(x)$ has been written as the Higgs field $h(x)$ to emphasise that this is the physical field in the unitary gauge. It is important to remember that the physical predictions of the theory do not depend on the choice of gauge, but in the unitary gauge the fields appearing in the Lagrangian correspond to the physical particles; there are no “mixing” terms like $B_\mu(\partial^\mu \xi)$. The degree of freedom

corresponding to the Goldstone field $\xi(x)$ no longer appears in the Lagrangian; it has been replaced by the degree of freedom corresponding to the longitudinal polarisation state of the now massive gauge field B . Sometimes it is said that the Goldstone boson has been “eaten” by the gauge field. Writing $\mu^2 = -\lambda v^2$, and working in the unitary gauge where $\phi(x) = \frac{1}{\sqrt{2}}(v + h(x))$, the Lagrangian of (17.28) can be written

$$\begin{aligned}\mathcal{L} &= (D_\mu \phi)^*(D^\mu \phi) - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \mu^2 \phi^2 - \lambda \phi^4 \\ &= \frac{1}{2} (\partial_\mu - ig B_\mu)(v + h)(\partial^\mu + ig B^\mu)(v + h) - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} \mu^2 (v + h)^2 - \frac{1}{4} \lambda (v + h)^4 \\ &= \frac{1}{2} (\partial_\mu h)(\partial^\mu h) + \frac{1}{2} g^2 B_\mu B^\mu (v + h)^2 - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \lambda v^2 h^2 - \lambda v h^3 - \frac{1}{4} \lambda h^4 + \frac{1}{4} \lambda v^4.\end{aligned}$$

Gathering up the terms (and ignoring the $\lambda v^4/4$ constant) gives

$$\begin{aligned}\mathcal{L} &= \underbrace{\frac{1}{2} (\partial_\mu h)(\partial^\mu h)}_{\text{massive } h \text{ scalar}} - \underbrace{\lambda v^2 h^2 - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} g^2 v^2 B_\mu B^\mu}_{\text{massive gauge boson}} \\ &\quad + \underbrace{g^2 v B_\mu B^\mu h + \frac{1}{2} g^2 B_\mu B^\mu h^2}_{h, B \text{ interactions}} - \underbrace{\lambda v h^3 - \frac{1}{4} \lambda h^4}_{h \text{ self-interactions}}.\end{aligned}\quad (17.34)$$

This Lagrangian describes a massive scalar Higgs field h and a massive gauge boson B associated with the $U(1)$ local gauge symmetry. It contains interaction terms between the Higgs boson and the gauge boson, and Higgs boson self-interaction terms, indicated in [Figure 17.11](#). The mass of the gauge boson,

$$m_B = g v,$$

is related to the strength of the gauge coupling and the vacuum expectation value of the Higgs field. The mass of the Higgs boson is given by

$$m_H = \sqrt{2\lambda} v.$$

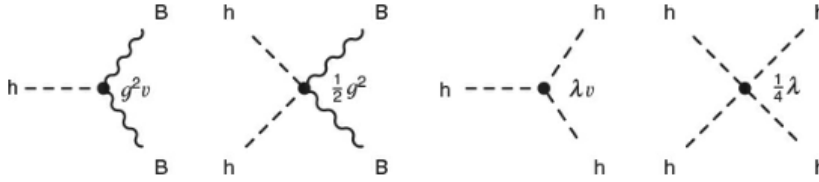


Fig. 17.11 The interaction terms arising from the Higgs mechanism for a U(1) local gauge theory.

It should be noted that the vacuum expectation value v sets the scale for the masses of both the gauge boson and the Higgs boson.

17.5.4 The Standard Model Higgs

In the above example, the Higgs mechanism was used to generate a mass for the gauge boson corresponding to a U(1) local gauge symmetry. In the Salam–Weinberg model, the Higgs mechanism is embedded in the $U(1)_Y \times SU(2)_L$ local gauge

symmetry of the electroweak sector of the Standard Model. Three Goldstone bosons will be required to provide the longitudinal degrees of freedom of the W^+ , W^- and Z bosons. In addition, after symmetry breaking, there will be (at least) one massive scalar particle corresponding to the field excitations in the direction picked out by the choice of the physical vacuum. The simplest Higgs model, which has the necessary four degrees of freedom, consists of two complex scalar fields.

Because the Higgs mechanism is required to generate the masses of the electroweak gauge bosons, one of the scalar fields must be neutral, written as ϕ^0 , and the other must be charged such that ϕ^+ and $(\phi^+)^* = \phi^-$ give the longitudinal degrees of freedom of the W^+ and W^- . The minimal Higgs model consists of two complex scalar fields, placed in a weak isospin doublet

$$\phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi_1 + i\phi_2 \\ \phi_3 + i\phi_4 \end{pmatrix}. \quad (17.35)$$

As usual, the upper and lower components of the doublet differ by one unit of charge. The Lagrangian for this doublet of complex scalar fields is

$$\mathcal{L} = (\partial_\mu \phi)^\dagger (\partial^\mu \phi) - V(\phi), \quad (17.36)$$

with the Higgs potential,

$$V(\phi) = \mu^2 \phi^\dagger \phi + \lambda (\phi^\dagger \phi)^2.$$

For $\mu^2 < 0$, the potential has an infinite set of degenerate minima satisfying

$$\phi^\dagger \phi = \frac{1}{2}(\phi_1^2 + \phi_2^2 + \phi_3^2 + \phi_4^2) = \frac{v^2}{2} = -\frac{\mu^2}{2\lambda}.$$

After symmetry breaking, the neutral photon is required to remain massless, and therefore the minimum of the potential must correspond to a non-zero vacuum expectation value only of the neutral scalar field ϕ^0 ,

$$\langle 0|\phi|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}.$$

The fields then can be expanded about this minimum by writing

$$\phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi_1(x) + i\phi_2(x) \\ v + \eta(x) + i\phi_4(x) \end{pmatrix}.$$

After the spontaneous breaking of the symmetry, there will be a massive scalar and three massless Goldstone bosons, which will ultimately give the longitudinal degrees of freedom of the $W^\pm$ and Z bosons. Rather than repeating the derivation given in [Section 17.5.3](#) and “gauging-away” the Goldstone fields, here the Higgs doublet is immediately written in the unitary gauge,

$$\phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}.$$

The resulting Lagrangian is known as the Salam–Weinberg model. All that remains is to identify the masses of gauge bosons and the interaction terms.

The mass terms can be identified by writing the Lagrangian of (17.36) such that it respects the $SU(2)_L \times U(1)_Y$ local gauge symmetry of the electroweak model by replacing the derivatives with the appropriate covariant derivatives (discussed further in [Appendix F](#)),

$$\partial_\mu \rightarrow D_\mu = \partial_\mu + ig_W \mathbf{T} \cdot \mathbf{W}_\mu + ig' \frac{Y}{2} B_\mu, \quad (17.37)$$

where $\mathbf{T} = \frac{1}{2} \boldsymbol{\sigma}$ are the three generators of the $SU(2)$ symmetry. In [Chapter 15](#), the weak hypercharge of the Glashow–Salam–Weinberg (GSW) model was identified as $Y = 2 \left(Q - I_W^{(3)} \right)$. Here, the lower component of the Higgs doublet is neutral and has $I_W^{(3)} = -\frac{1}{2}$ and thus the Higgs doublet has hypercharge $Y = 1$.

Hence, the effect of the covariant derivative of (17.37) acting on the Higgs doublet ϕ is

$$D_\mu \phi = \frac{1}{2} \left[2\partial_\mu + \left(ig_W \boldsymbol{\sigma} \cdot \mathbf{W}_\mu + ig' B_\mu \right) \right] \phi,$$

where D_μ is a 2×2 matrix acting on the two component weak isospin doublet and the identity matrix multiplying the ∂_μ and B_μ terms is implicit in this expression.

The term in the Lagrangian that generates the masses of the gauge bosons is $(D_\mu \phi)^\dagger (D^\mu \phi)$. In the Unitary gauge $D_\mu \phi$ is given by

$$\begin{aligned} D_\mu \phi &= \frac{1}{2\sqrt{2}} \begin{pmatrix} 2\partial_\mu + ig_W W_\mu^{(3)} + ig' B_\mu & ig_W [W_\mu^{(1)} - iW_\mu^{(2)}] \\ ig_W [W_\mu^{(1)} + iW_\mu^{(2)}] & 2\partial_\mu - ig_W W_\mu^{(3)} + ig' B_\mu \end{pmatrix} \begin{pmatrix} 0 \\ v + h \end{pmatrix} \\ &= \frac{1}{2\sqrt{2}} \begin{pmatrix} ig_W (W_\mu^{(1)} - iW_\mu^{(2)})(v + h) \\ (2\partial_\mu - ig_W W_\mu^{(3)} + ig' B_\mu)(v + h) \end{pmatrix}. \end{aligned}$$

Taking the Hermitian conjugate gives $(D_\mu \phi)^\dagger$, from which

$$(D_\mu \phi)^\dagger (D^\mu \phi) = \frac{1}{2}(\partial_\mu h)(\partial^\mu h) + \frac{1}{8}g_W^2(W_\mu^{(1)} + iW_\mu^{(2)})(W^{(1)\mu} - iW^{(2)\mu})(v+h)^2 \\ + \frac{1}{8}(g_W W_\mu^{(3)} - g' B_\mu)(g_W W^{(3)\mu} - g' B^\mu)(v+h)^2. \quad (17.38)$$

The gauge bosons masses are determined by the terms in $(D_\mu \square)^\dagger (D^\mu \square)$ that are quadratic in the gauge boson fields, i.e.

$$\frac{1}{8}v^2 g_W^2 (W_\mu^{(1)} W^{(1)\mu} + W_\mu^{(2)} W^{(2)\mu}) + \frac{1}{8}v^2 (g_W W_\mu^{(3)} - g' B_\mu)(g_W W^{(3)\mu} - g' B^\mu).$$

In the Lagrangian, the mass terms for the $W^{(1)}$ and $W^{(2)}$ spin-1 fields will appear as

$$\frac{1}{2}m_W^2 W_\mu^{(1)} W^{(1)\mu} \quad \text{and} \quad \frac{1}{2}m_W^2 W_\mu^{(2)} W^{(2)\mu},$$

and therefore the mass of the W boson is

$$m_W = \frac{1}{2}g_W v. \quad (17.39)$$

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which are related to the vacuum expectation value of the Higgs field v and the mass of the Higgs boson m_H by

$$v^2 = \frac{-\mu^2}{\lambda} \quad \text{and} \quad m_H^2 = 2\lambda v^2.$$

By using the relation $m_W = \frac{1}{2}g_W v$ and the measured values for m_W and g_W , the vacuum expectation value of the Higgs field is found to be

$$v = 246 \text{ GeV}.$$

The parameter λ in the Higgs potential can be obtained from the mass of the Higgs boson as measured at the LHC (see [Section 17.7](#)).

The rest of the this and next chapter continues to show the mass generation of the other weak bosons, e.g. Z it can be found elsewhere skip the rest – the big reason the Higg's is first introduced but then a similar mechanism is applied to all particles, picking up there

I include one page from in the skipped range since this textbook is more recent, compared to Griff which was before the Higg's mass was known

Hence the coupling of the Higgs boson to the W boson is proportional to the W-boson mass. Likewise, the coupling of the Higgs boson to the Z boson, $g_{HZZ} = g_Z m_Z$, is proportional to m_Z .

17.5.5 Fermion masses

The Higgs mechanism for the spontaneous symmetry breaking of the $U(1)_Y \times SU(2)_L$ gauge group of the Standard Model generates the masses of the W and Z bosons. Remarkably, it also can be used to generate the masses of the fermions. Because of the different transformation properties of left- and right-handed chiral states, the fermion mass term in the Dirac Lagrangian,

$$-m\bar{\psi}\psi = -m(\bar{\psi}_R\psi_L + \bar{\psi}_L\psi_R),$$

does not respect the $SU(2)_L \times U(1)_Y$ gauge symmetry, and therefore cannot be present in the Lagrangian of the Standard Model.

In the Standard Model, left-handed chiral fermions are placed in $SU(2)$ doublets, here written L , and right-handed fermions are placed in $SU(2)$ singlets, here denoted R . Because the two complex scalar fields of the Higgs mechanism are placed in an $SU(2)$ doublet $\Phi(x)$, an infinitesimal $SU(2)$ local gauge transformation has the effect,

$$\phi \rightarrow \phi' = (I + ig_W \epsilon(x) \cdot \mathbf{T})\phi.$$

Exactly the same local gauge transformation applies to the left-handed doublet of fermion fields L . Therefore, the effect of the infinitesimal $SU(2)$ gauge transformation on $\bar{L} \equiv L^\dagger \gamma^0$ is

$$\bar{L} \rightarrow \bar{L}' = \bar{L}(I - ig_W \epsilon(x) \cdot \mathbf{T}).$$

Consequently, the combination $\bar{L}\phi$ is invariant under the $SU(2)_L$ gauge transformations. When combined with a right-handed singlet, $\bar{L}\phi R$, it is invariant under $SU(2)_L$ and $U(1)_Y$ gauge transformations; as is its Hermitian

conjugate $(\bar{L}\phi R)^\dagger = \bar{R}\phi^\dagger L$. Hence, a term in the Lagrangian of the form $-g_f(\bar{L}\phi R + \bar{R}\phi^\dagger L)$ satisfies the $SU(2)_L \times U(1)_Y$ gauge symmetry of the Standard Model. For the $SU(2)_L$ doublet containing the electron, this corresponds to

$$\mathcal{L}_e = -g_e \left[\begin{pmatrix} \bar{\nu}_e & \bar{e} \end{pmatrix}_L \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} e_R + \bar{e}_R \begin{pmatrix} \phi^{+*} & \phi^{0*} \end{pmatrix} \begin{pmatrix} \nu_e \\ e \end{pmatrix}_L \right], \quad (17.46)$$

where g_e is a constant known as the Yukawa coupling of the electron to the Higgs field. After spontaneously symmetry breaking, the Higgs doublet in the unitary gauge is

$$\phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix},$$

and thus (17.46) becomes

$$\mathcal{L}_e = -\frac{g_e}{\sqrt{2}} v (\bar{e}_L e_R + \bar{e}_R e_L) - \frac{g_e}{\sqrt{2}} h (\bar{e}_L e_R + \bar{e}_R e_L). \quad (17.47)$$

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The first term in (17.47) has exactly the form required for the fermion masses, but has now been introduced in gauge invariant manner. The Yukawa coupling g_e is not predicted by the Higgs mechanism, but can be chosen to be consistent with the observed electron mass,

$$g_e = \sqrt{2} \frac{m_e}{v}.$$

In this case, (17.47) becomes

$$\mathcal{L}_e = -m_e \bar{e} e - \frac{m_e}{v} \bar{e} e h. \quad (17.48)$$

The first term in (17.48), which gives the mass of the electron, represents the coupling of electron to the Higgs *field* through its non-zero vacuum expectation value. The second term in (17.48) gives rise to a coupling between the electron and the Higgs *boson* itself. These two terms are illustrated in Figure 17.13, where the fermion masses arise from the coupling of left-handed and right-handed

massless chiral fermions though the interaction with the non-zero expectation value of the Higgs field.

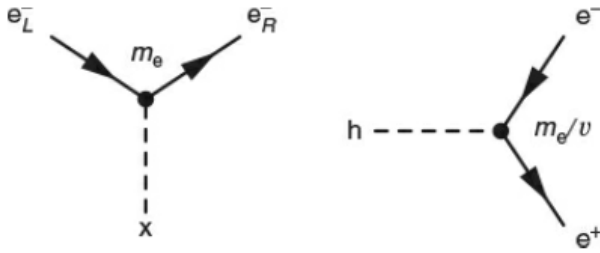


Fig. 17.13 Left: the interaction between a massless chiral electron and the non-zero expectation value of the Higgs field. Right: the interaction vertex for the coupling of the Higgs boson to an electron.

Because the non-zero vacuum expectation value occurs in the lower (neutral) component of the Higgs doublet, the combination of fields $\bar{L}\phi R + \bar{R}\phi^\dagger L$ only can generate the masses for the fermion in the lower component of an $SU(2)_L$ doublet. Thus it can be used to generate the masses of the charged leptons and the down-type quarks, but not the up-type quarks or the neutrinos. Putting aside

the question of neutrino masses, a mechanism is required to give masses to the up-type quarks. This can be achieved by constructing the conjugate doublet $\bar{\Phi}_c$ formed from the four fields in (17.35),

$$\bar{\Phi}_c = -i\sigma_2\Phi^* = \begin{pmatrix} -\phi^{0*} \\ \phi^- \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -\phi_3 + i\phi_4 \\ \phi_1 - i\phi_2 \end{pmatrix}.$$

Because of the particular properties of $SU(2)$, see Section 9.5, the conjugate doublet $\bar{\Phi}_c$ transforms in exactly the same way as the doublet Φ . This is analogous to the representation of up- and down-quarks and anti-up and anti-down in $SU(2)$ isospin symmetry. A gauge invariant mass term for the up-type quarks can be constructed from $\bar{L}\phi_c R + \bar{R}\phi_c^\dagger L$, for example

$$\mathcal{L}_u = g_u (\bar{u} \ \bar{d})_L \begin{pmatrix} -\phi^{0*} \\ \phi^- \end{pmatrix} u_R + \text{Hermitian conjugate},$$

which after symmetry breaking becomes

$$\mathcal{L}_u = -\frac{g_u}{\sqrt{2}}v(\bar{u}_L u_R + \bar{u}_R u_L) - \frac{g_u}{\sqrt{2}}h(\bar{u}_L u_R + \bar{u}_R u_L),$$

with the Yukawa coupling $g_u = \sqrt{2}m_u/v$, giving

$$\mathcal{L}_u = -m_u \bar{u}u - \frac{m_u}{v} \bar{u}u h.$$

Hence for all *Dirac* fermions, gauge invariant mass terms can be constructed from either

$$\mathcal{L} = -g_f [\bar{L}\phi R + (\bar{L}\phi R)^\dagger] \quad \text{or} \quad \mathcal{L} = g_f [\bar{L}\phi_c R + (\bar{L}\phi_c R)^\dagger],$$

giving rise to both the masses of the fermions and the interactions between the Higgs boson and the fermion. The Yukawa couplings of the fermions to the Higgs field are given by

$$g_f = \sqrt{2}\frac{m_f}{v},$$

where the vacuum expectation value of the Higgs field is $v = 246$ GeV. Interestingly, for the top quark with $m_t \sim 173.5 \pm 1.0$ GeV, the Yukawa coupling is almost exactly unity. Whilst this may be a coincidence, it is perhaps natural that the Yukawa couplings of the fermions are $\mathcal{O}(1)$. If the neutrino masses are also associated with the Higgs mechanism, it is perhaps surprising that they are so small, with Yukawa couplings of $\lesssim 10^{-12}$. This might suggest that the mechanism which generates the neutrino masses differs from that for other fermions. One interesting possibility is the seesaw mechanism described in the addendum to this chapter.

17.8 *Addendum: Neutrinomasses

The right-handed chiral neutrino states ν_R do not participate in any of the interactions of the Standard Model; they do not couple to the gluons or electroweak gauge bosons. Consequently, there is no direct evidence that they exist. However, from the studies of neutrino oscillations it is known that neutrinos do have mass, and therefore there must be a corresponding mass term in the Lagrangian. In the Standard Model, neutrino masses can be introduced in exactly the same way as for the up-type quarks using the conjugate Higgs doublet. In this case, after spontaneous symmetry breaking, the gauge invariant *Dirac mass* term for the neutrino is

$$\mathcal{L}_D = -m_D (\bar{\nu}_R \nu_L + \bar{\nu}_L \nu_R).$$

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If this is the origin of neutrino masses, then right-handed chiral neutrinos exist. However, the neutrino masses are very much smaller than the masses of the other fermions, suggesting that another mechanism for generating neutrino mass might be present.

Because the *right-handed neutrinos* and *left-handed antineutrinos* transform as singlets under the Standard Model gauge transformations, any additional terms in the Lagrangian formed from these fields alone can be added to the Lagrangian without breaking the gauge invariance of the Standard Model. The left-handed antineutrinos appear in the Lagrangian as the CP conjugate fields defined by

$$\psi^c = \hat{C} \hat{P} \psi = i\gamma^2 \gamma^0 \psi^*,$$

where the CP conjugate field for the right-handed neutrino, written ν_R^c , corresponds to a left-handed antineutrino. Therefore the Majorana mass term

$$\mathcal{L}_M = -\frac{1}{2} M (\bar{\nu}_R^c \nu_R + \bar{\nu}_R \nu_R^c),$$

which is formed from right-handed neutrino fields and left-handed antineutrino fields, respects the local gauge invariance of the Standard Model. Consequently, it can be added to the Standard Model Lagrangian. However, there is a price to pay; the Majorana mass term provides a direct coupling between between a particle and an antiparticle. For example, the corresponding Majorana mass term for the electron would allow $e^+ \leftrightarrow e^-$ transitions, violating charge conservation. This problem does not exist for the neutrinos. Furthermore, because neutrinos are neutral, it is possible that they are their own antiparticles, in which case they are referred to as *Majorana neutrinos* as opposed to Dirac neutrinos.

17.8.1 The seesaw mechanism

The most general renormalisable Lagrangian for the neutrino masses includes both the Dirac and Majorana mass terms, indicated in [Figure 17.20](#). Because $\overline{\nu}_L \nu_R$ is equivalent to $\overline{\nu}_R^c \nu_L^c$, the Dirac mass term can be written

$$\mathcal{L}_D = -\frac{1}{2} m_D (\overline{\nu}_L \nu_R + \overline{\nu}_R^c \nu_L^c) + h.c.,$$

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where *h.c.* stands for the corresponding Hermitian conjugate. This term admits the possibility that neutrino masses arise from the spontaneous symmetry breaking of the Higgs mechanism. If in addition, the automatically gauge-invariant Majorana mass term is added by hand, the Lagrangian for the combined Dirac and Majorana masses is

$$\mathcal{L}_{DM} = -\frac{1}{2} \left[m_D \overline{\nu}_L \nu_R + m_D \overline{\nu}_R^c \nu_L^c + M \overline{\nu}_R^c \nu_R \right] + h.c.$$

or, equivalently,

$$\mathcal{L}_{DM} = -\frac{1}{2} \begin{pmatrix} \overline{\nu}_L & \overline{\nu}_R^c \end{pmatrix} \begin{pmatrix} 0 & m_D \\ m_D & M \end{pmatrix} \begin{pmatrix} \nu_L^c \\ \nu_R \end{pmatrix} + h.c. \quad (17.53)$$

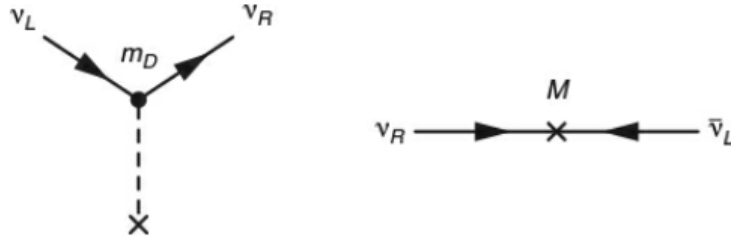


Fig. 17.20 The Dirac and Majorana neutrino mass terms.

The physical states of this system can be obtained from the basis in which the mass matrix is diagonal, analogous to the procedure for identifying the physical states of the neutral kaon system and the neutral gauge bosons of the $U(1)_Y \times SU(2)_L$ gauge symmetry. Hence, with the Lagrangian including Dirac and Majorana mass terms, the masses of the physical neutrino states are the eigenvalues of mass matrix $\mathbf{M}$ in (17.53). These can be found from the characteristic equation $\det(\mathbf{M} - \lambda I) = 0$, which implies $\lambda^2 - M\lambda - m_D^2 = 0$. Hence, in this model, the masses of the physical neutrinos would be

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$$m_{\pm} = \lambda_{\pm} = \frac{M \pm \sqrt{M^2 + 4m_D^2}}{2} = \frac{M \pm M \sqrt{1 + 4m_D^2/M^2}}{2}.$$

If the Majorana mass M is taken to be much greater than the Dirac mass m_D , then

$$m_{\pm} \approx \frac{1}{2}M \pm \frac{1}{2} \left(M + \frac{2m_D^2}{M} \right), \quad (17.54)$$

giving a light neutrino state² (ν) and heavy neutrino state (N) with masses

$$|m_{\nu}| \approx \frac{m_D^2}{M} \quad \text{and} \quad m_N \approx M.$$

In the seesaw mechanism, it is hypothesised that the Dirac mass terms for the neutrinos are of a similar size to the masses of the other fermions, i.e. $\mathcal{O}(1 \text{ GeV})$.

The Majorana mass M is then made sufficiently large that the lighter of the two physical neutrino states has a mass $m_\nu \sim 0.01$ eV. In this way, the masses of the lighter neutrino states can be made to be very small, even when the Dirac mass term is of the same order of magnitude as the other fermions. For this to work, the Majorana mass must be very large, $M \sim 10^{11}$ GeV.

If a Majorana mass term exists, the seesaw mechanism predicts that for each of the three neutrino generations, there is a very light neutrino with a mass much smaller than the other Standard Model fermions and a very massive neutrino state $m_N \approx M$. The physical neutrino states, which are obtained from the eigenvalues of the mass matrix, are

$$\nu = \cos \theta (\nu_L + \nu_L^c) - \sin \theta (\nu_R + \nu_R^c) \quad \text{and} \quad N = \cos \theta (\nu_R + \nu_R^c) + \sin \theta (\nu_L + \nu_L^c),$$

where $\tan \theta \approx m_D/M$. Since the left-handed chiral components of the light neutrino are multiplied by $\cos \theta$, the effect of introducing the Majorana mass term is to reduce the weak charged-current couplings of the light neutrino states by a factor $\cos \theta$. However, for $M \gg m_D$, the neutrino states are

$$\nu \approx (\nu_L + \nu_L^c) - \frac{m_D}{M} (\nu_R + \nu_R^c) \quad \text{and} \quad N \approx (\nu_R + \nu_R^c) + \frac{m_D}{M} (\nu_L + \nu_L^c),$$

and the couplings of the light neutrinos to the weak charged-current are essentially the same as those of the Standard Model. Since the massive neutrino state is almost entirely right-handed, it would not participate in the weak charged- or neutral-currents.

The seesaw mechanism provides an interesting hypothesis for the smallness of neutrino masses, but it is just a hypothesis. It would be placed on firmer ground if neutrinos were shown to be Majorana particles. One experimental consequence of neutrinos being Majorana particles would be the possibility of observing the phenomenon of neutrinoless double β -decay, which is discussed in the following chapter.

Problems